CEN-505 Roll No. 20BCS021

B. Tech. (Computer Engineering)
V Semester Examination- 2022
COMPUTER NETWORKS
Paper No.: CEN-505

Time: 3 Hours **MM: 60**

Note: Attempt all questions. Make suitable assumption(s) wherever required.

Q. 1 CO1 (a) Discuss the following with respect to each- physical layer, data link layer, network layer and transport layer of OSI model: **(6)**

- Name of the data unit
- Addressing Scheme used
- Device used
- Delivery of data (from and to)

(b)(i) A TCP message consisting of 2100 bytes is passed to IP for delivery across two networks. The first network can carry a maximum payload of 1200 bytes per frame and the second network can carry a maximum payload of 400 bytes per frame, excluding network overhead. Assume that IP overhead per packet is 20 bytes. What is the total IP overhead in the second network for this transmission? **(3+3)**

(ii) The transport layer protocols used for real time multimedia, file transfer, DNS and email, respectively are
(A) TCP, UDP, UDP and TCP
(B) UDP, TCP, TCP and UDP
(C) UDP, TCP, UDP and TCP
(D) TCP, UDP, TCP and UDP
Justify your answer.

OR

(b') An ISP is granted a block of addresses starting with 190.100.0.0/16. The ISP needs to distribute these addresses to three groups of customers as follows: **(6)**
a. The first group has 32 customers; each needs 512 addresses.
b. The second group has 256 customers; each needs 64 addresses
c. The third group has 128 customers; each needs 32 addresses.
Design the sub blocks and find out how many addresses are still available after these allocations.

Q. 2 CO2 (a) Consider the store and forward packet switched network given below. Assume that the bandwidth of each link is 10^6 bytes / sec. A user on host A sends a file of size 10^3 bytes to host B through routers R1 and R2 in three different ways. In the first case a single packet containing the complete file is transmitted from A to B. In the second case, the file is split into 10 equal parts, and these packets are transmitted from A to B. In the third case, the file is split into 20 equal parts and these packets are sent from A to B. Each packet contains 100 bytes of header information along with the user data. Consider only transmission time and ignore processing, queuing and propagation delays. Also assume that there are no errors during transmission. Calculate the times taken to transmit the file in the first, second and third case respectively. **(6)**

(b) Differentiate guided and unguided transmission media. Discuss various bands in unguided transmission media?

OR

(b')(i) Discuss how packet switching can be connection oriented? In what way, it is different from circuit switching. (3

(ii) Comment on the following statements with proper justification:
(A) Packet switching leads to better utilization of bandwidth resources than circuit Switching.
(B) Packet switching results in less variation in delay than circuit switching.
(C) Packet switching requires more per packet processing than circuit switching.
(D) Packet switching can lead to reordering unlike in circuit switching.

Q. 3 **(a)(i)** A 1 Mbps satellite link connects two ground stations. The altitude of the satellite is 36,504 Km and speed of the signal is 3*10^8 m/sec. What should be the packet size for a channel utilization of 25% for a satellite link using Go- Back-127 sliding window protocol? **(3+3)**

CO3

(ii) Consider a selective repeat sliding window protocol that uses a frame size of 1 KB to send data on a 1.5 Mbps link with a one-way latency of 50 msec. To achieve a link utilization of 60%, calculate the minimum number of bits required to represent the sequence number field?

(b)(i) A bit stuffing based framing protocol uses a 5-bit delimiter pattern of 01110. If the output bit string after stuffing is 01100110, determine the input bit string? **(3+3)**

(ii) Consider the cyclic redundancy check (CRC) based error detecting scheme having the generator polynomial x^3+x+1. Suppose the message 11000 is to be transmitted. Check bits are appended at the end of the message by the transmitter using the above CRC scheme. Calculate the transmitted bit sequence.

OR

(b') Discuss why the size of the sending window in Go-Back-N ARQ is (2^n)-1, while in the case of selective repeat it is 2^ (n-1). Justify your answer by giving proper example. **(6)**

Q. 4 **(a)(i)** Define CSMA/CD (carrier sense multiple access / collision detection). **(3+3)**

CO4 (ii) Consider a CSMA/CD network that transmits data at a rate of 100 Mbps (10^8 bits per second) over a 1 km cable with no repeaters. If the minimum frame size required for this network is 1250 bytes, what is the signal speed (km/sec) in the cable?

(b)(i) Draw the frame structure of 802.11 for wireless LAN. **(3+3)**

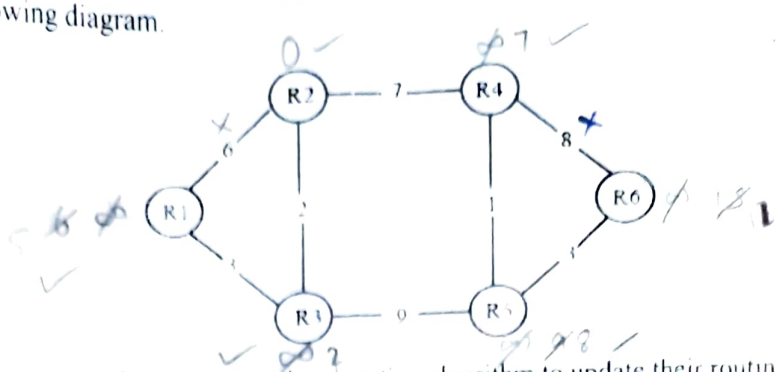
(ii) In wireless LAN, the point coordination function and distributed coordination function can coexist within one cell. Explain how it is possible to have central control and distributed control operating at the same time.

OR

(b')(i) What is piconet? Bluetooth device can be in two piconets at the same time. Is there any reason why one device cannot be the master in both of them at the same time? **(3+3)**

(ii) A group of N stations share a 50-kbps slotted aloha channel. Each station output a 500 bits frame on an average of once every 5000 msec, even if previous one has not been sent. What is the maximum value of N?

(a) Consider a network with 6 routers R1 to R6 connected with links having weights as shown in the following diagram. **(6)**



All routers used the distance vector base routing algorithm to update their routing tables. Each router starts with its routing table initialized to contain an entry for each neighbor with the weight of the respective connecting link. After all the routing tables stabilize, show how many links in the network will never be used for carrying any data?

(b)(i) What are the benefits of token bucket over leaky bucket, explain with suitable example? **(3+3)**

(ii) A computer on a 10 Mbps network is regulated by a token bucket. The token bucket is filled at a rate of 2 Mbps. It is initially filled to capacity with 16 Megabits. What is the maximum duration for which the computer can transmit at the full 10 Mbps? Also, find the total duration it will take to transmit the complete 32 Megabits input burst?

OR

(b') For the network given below, obtain the routing table of router 'B' using link state routing. **(6)**

